

Tracing bone marrow-derived microglia in brain metastatic tumors.

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Microglia are the resident macrophages in the central nervous system (CNS), and they constitute 15-20% of the total glial populations. They have wide developmental and protective functions during brain injury, infection and tumorigenesis. Originally thought to derive from postnatal hematopoietic progenitors, it has recently been demonstrated that microglia originate from primitive myeloid progenitor cells that arise during early development from the embryonic yolk sac. Circulating monocytes infiltrate the CNS upon inflammatory conditions, such as cancer, primarily differentiating into macrophages and dendritic cells. Both resident and recruited microglia respond to environmental cues and actively participate in pathogenic processes, albeit their transcriptomic profiles contain significant differences suggesting distinctive roles. Metastatic brain tumors are the most common intracranial neoplasm in adults, with an estimate incidence 10 times higher than all primary brain neoplasms combined, and with dismal prognosis. Microglia is a major immune population associated with brain metastatic tumors in patients. They are proposed to play multiple, and sometimes opposing roles, in tumor progression. However, our ability to evaluate individual contribution of resident and recruited populations is hindered by the fact that they express overlapping sets of surface markers. Tracking and interrogating tissue-resident vs recruited microglia in the brain tumor microenvironment becomes critical to dissect their respective roles and gain a better understanding of the mechanism governing their interaction. In this chapter, we describe the utilization of genetic reporter mice to identify recruited brain microglia, offer a comparison between the genetic method and the most widely used flow cytometric approach, and discuss potential downstream applications to interrogate BMDM function in brain metastatic disease.

Microglia - the brain's busy bees.

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Recent years have seen significant changes in the way scientists view microglia and their role in health and disease. For decades, it was presumed that microglia were stationary, inactive immune cells in the brain, waiting for an immunologic insult to call them into action. In contrast, modern imaging techniques have revealed that microglia are constantly in motion, surveying their environment. Lineage tracing studies have led to the understanding that microglia are part of a larger family of cells, called tissue-resident macrophages, which arise from early yolk sac progenitors during embryogenesis and engraft nearly every organ in the body. Microglia, and all tissue-resident macrophages, rely on signaling through CD115 (the colony stimulating factor 1 receptor) for survival, primarily through the ligand, macrophage colony-stimulating factor. However, it is now understood that some microglia have a specific need for another CD115 ligand, Interleukin-34, which is only shared with Langerhans cells in the skin. In contrast to classical views, recent evidence suggests that the primary functions of microglia may occur during postnatal neurodevelopment and adult homeostasis, as absence or impairment of microglia results in a pathology separate from inflammatory immune function. In summary, these advances suggest that microglia might eventually be utilized or targeted to improve disease outcomes via encouraging or enhancing their health-promoting homeostatic functions.

Glial Cells Promote Myelin Formation and Elimination.

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Building a functional nervous system requires the coordinated actions of many glial cells. In the vertebrate central nervous system (CNS), oligodendrocytes myelinate neuronal axons to increase conduction velocity and provide trophic support. Myelination can be modified by local signaling at the axon-myelin interface, potentially adapting sheaths to support the metabolic needs and physiology of individual neurons. However, neurons and oligodendrocytes are not wholly responsible for crafting the myelination patterns seen in vivo. Other cell types of the CNS, including microglia and astrocytes, modify myelination. In this review, I cover the contributions of non-neuronal, non-oligodendroglial cells to the formation, maintenance, and pruning of myelin sheaths. I address ways that these cell types interact with the oligodendrocyte lineage throughout development to modify myelination. Additionally, I discuss mechanisms by which these cells may indirectly tune myelination by regulating neuronal activity. Understanding how glial-glial interactions regulate myelination is essential for understanding how the brain functions as a whole and for developing strategies to repair myelin in disease.

Glial lineages and myelination in the central nervous system.

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Oligodendrocytes, derived from stem cell precursors which arise in subventricular zones of the developing central nervous system, have as their specialist role the synthesis and maintenance of myelin. Astrocytes contribute to the cellular architecture of the central nervous system and act as a source of growth factors and cytokines; microglia are bone-marrow derived macrophages which function as primary immunocompetent cells in the central nervous system. Myelination depends on the establishment of stable relationships between each differentiated oligodendrocyte and short segments of several neighbouring axons. There is growing evidence, especially from studies of glial cell implantation, that oligodendrocyte precursors persist in the adult nervous system and provide a limited capacity for the restoration of structure and function in myelinated pathways damaged by injury or disease.

Enriched Environment Effects on Myelination of the Central Nervous System: Role of Glial Cells.

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Myelination is regulated by various glial cells in the central nervous system (CNS), including oligodendrocytes (OLs), microglia, and astrocytes. Myelination of the CNS requires the generation of functionally mature OLs from OPCs. OLs are the myelin-forming cells in the CNS. Microglia play both beneficial and detrimental roles during myelin damage and repair. Astrocyte is responsible for myelin formation and regeneration by direct interaction with oligodendrocyte lineage cells. These glial cells are influenced by experience-dependent activities such as environmental enrichment (EE). To date, there are few studies that have investigated the association between EE and glial cells. EE with a complex combination of sensorimotor, cognitive, and social stimulation has a significant effect on cognitive impairment and brain plasticity. Hence, one mechanism through EE improving cognitive function may rely on the mutual effect of EE and glial cells. The purpose of this paper is to review recent research into the efficacy of EE for myelination and glial cells at cellular and molecular levels and offers critical insights for future research directions of EE and the treatment of EE in cognitive impairment disease.

[Myelination, demyelination and re-myelination in the central nervous system].

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The myelin sheaths that surround axons in the CNS are made and maintained by oligodendrocytes. These glial cells can form variable numbers of myelin segments (internodes): from 1 to 200 so that when one oligodendrocyte is destroyed with preservation of the axon, many internodes can be lost, constituting a demyelinating process. As a consequence of the destruction of myelin and sheath cells a rapid and abundant cell response takes place. The response is made up by resident (microglia) and haematogenous phagocytes which phagocytose myelin and cellular debris leaving the axons demyelinated. Demyelinated axons may either stay demyelinated and clumped together or they may be separated by astrocytic processes, yet they can be remyelinated. The occurrence of remyelination depends upon the intensity and time of exposition to the demyelinating agent. Remyelination in the CNS with complete restoration of conduction may be made by oligodendrocytes or Schwann cells which invade the CNS when astrocytes are destroyed.

Molecular mechanisms regulating myelination in the peripheral nervous system.

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Glial cells and neurons are engaged in a continuous and highly regulated bidirectional dialog. A remarkable example is the control of myelination. Oligodendrocytes in the central nervous system (CNS) and Schwann cells (SCs) in the peripheral nervous system (PNS) wrap their plasma membranes around axons to organize myelinated nerve fibers that allow rapid saltatory conduction. The functionality of this system is critical, as revealed by numerous neurological diseases that result from deregulation of the system, including multiple sclerosis and peripheral neuropathies. In this review we focus on PNS myelination and present a conceptual framework that integrates crucial signaling mechanisms with basic SC biology. We will highlight signaling hubs and overarching molecular mechanisms, including genetic, epigenetic, and post-translational controls, which together regulate the interplay between SCs and axons, extracellular signals, and the transcriptional network.

Myelination of the nervous system: mechanisms and functions.

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Myelination of axons in the nervous system of vertebrates enables fast, saltatory impulse propagation, one of the best-understood concepts in neurophysiology. However, it took a long while to recognize the mechanistic complexity both of myelination by oligodendrocytes and Schwann cells and of their cellular interactions. In this review, we highlight recent advances in our understanding of myelin biogenesis, its lifelong plasticity, and the reciprocal interactions of myelinating glia with the axons they ensheath. In the central nervous system, myelination is also stimulated by axonal activity and astrocytes, whereas myelin clearance involves microglia/macrophages. Once myelinated, the long-term integrity of axons depends on glial supply of metabolites and neurotrophic factors. The relevance of this axoglial symbiosis is illustrated in normal brain aging and human myelin diseases, which can be studied in corresponding mouse models. Thus, myelinating cells serve a key role in preserving the connectivity and functions of a

healthy nervous system.

Advances in myelinating glial cell development.

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In the vertebrate nervous system, the fast conduction of action potentials is potentiated by the myelin sheath, a multi-lamellar, lipid-rich structure that also provides vital trophic and metabolic support to axons. Myelin is elaborated by the plasma membrane of specialized glial cells, oligodendrocytes in the central nervous system (CNS) and Schwann cells (SCs) in the peripheral nervous system (PNS). The diseases that result from damage to myelin or glia, including multiple sclerosis and Charcot-Marie-Tooth disease, underscore the importance of these cells for human health. Therefore, an understanding of glial development and myelination is crucial in addressing the etiology of demyelinating diseases and developing patient therapies. In this review, we discuss new insights into the roles of mechanotransduction and cytoskeletal rearrangements as well as activity dependent myelination and axonal maintenance by glia. Together, these discoveries advance our knowledge of myelin and glia in nervous system health and plasticity throughout life.

Motor Exit Point (MEP) Glia: Novel Myelinating Glia That Bridge CNS and PNS Myelin.

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Oligodendrocytes (OLs) and Schwann cells (SCs) have traditionally been thought of as the exclusive myelinating glial cells of the central and peripheral nervous systems (CNS and PNS), respectively, for a little over a century. However, recent studies demonstrate the existence of a novel, centrally-derived peripheral glial population called motor exit point (MEP) glia, which myelinate spinal motor root axons in the periphery. Until recently, the boundaries that exist between the CNS and PNS, and the cells permitted to cross them, were mostly described based on fixed histological collections and static lineage tracing. Recent work in zebrafish using in vivo, time-lapse imaging has shed light on glial cell interactions at the MEP transition zone and reveals a more complex picture of myelination both centrally and peripherally.
